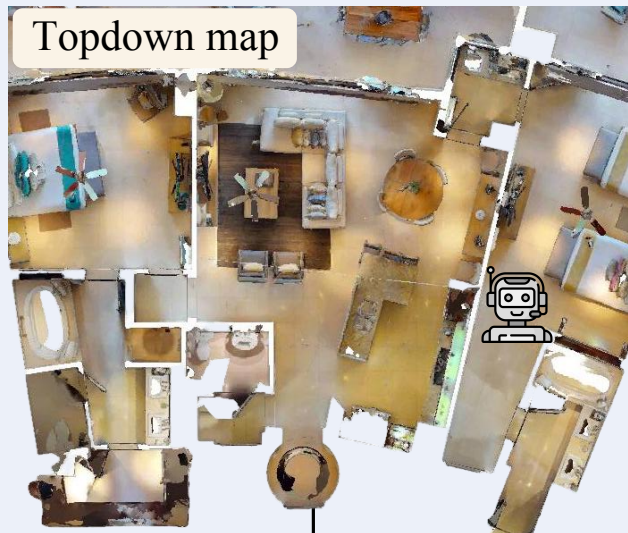


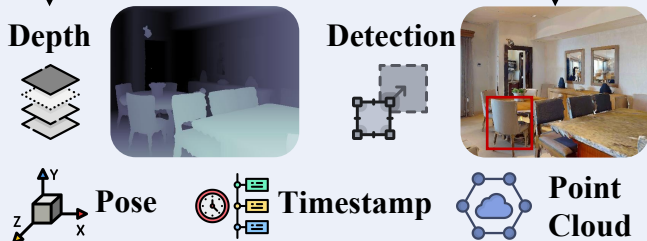
(a) Causal Evidence

Scene Observations



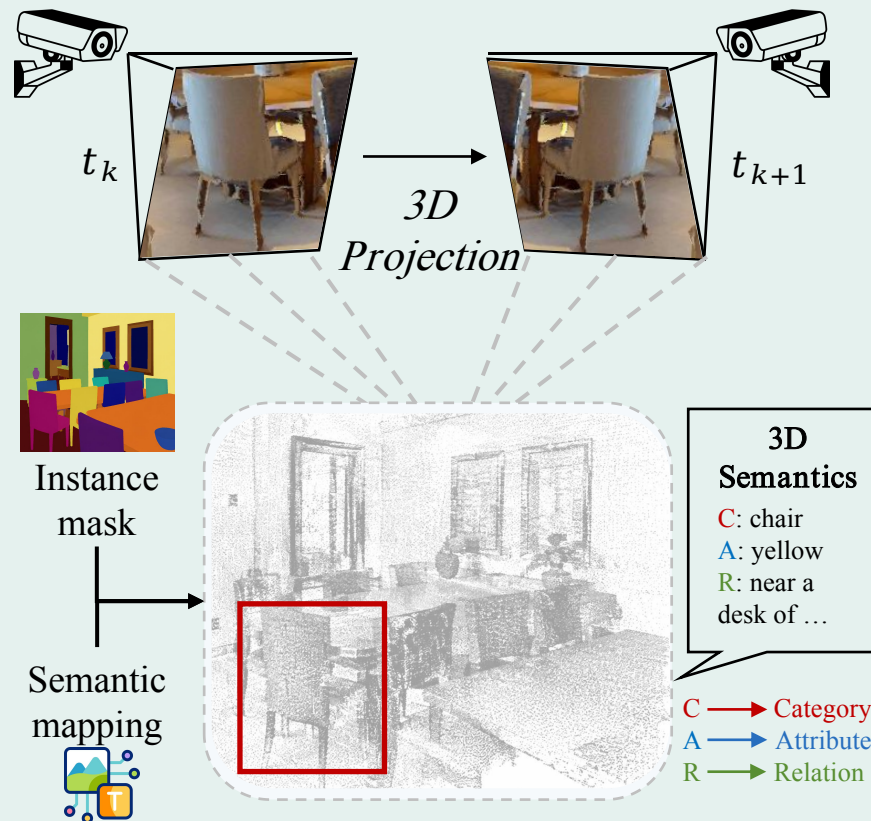
Online Observation Packet

Grounding DINO SAM 2

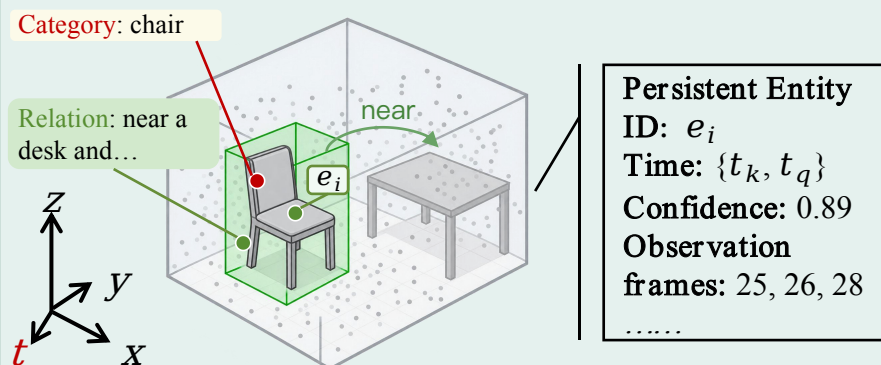


(b) Entity-Centric 3D Grounding

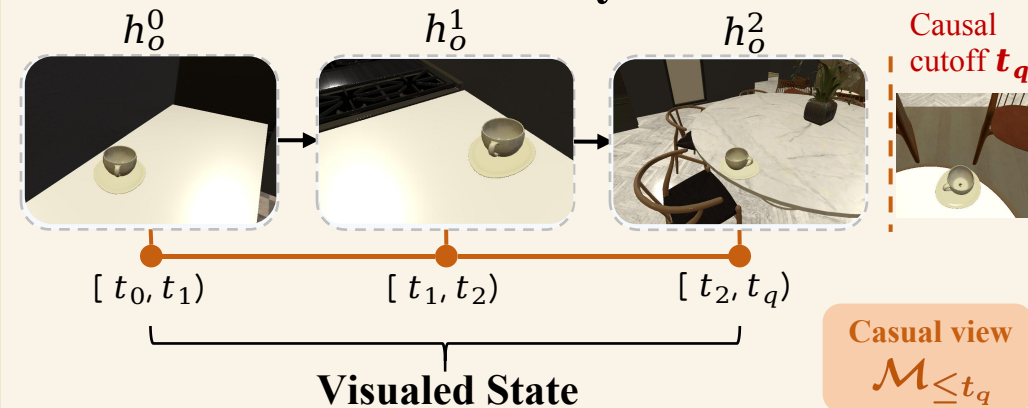
Multi-view observations



Persistent Grounded Entity



(c) Persistent 4D Memory

Semantic-Spatial Graph $\mathcal{G}_t$ Time-Valid Entity Versions $\mathcal{V}_t$ Evidence Provenance $\mathcal{B}_t$ Provenance of version v_i^2 (e_i at $[t_2, t_q]$)

$\mathcal{M}_{\leq t_q} \rightarrow$ Retrieve entity history $\mathcal{H}_t \rightarrow$ Predictive 4D Belief